

Letter

Accelerating Moment Tensor Potentials through post-training pruning

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ABSTRACT

Moment Tensor Potentials (MTPs) are machine-learning interatomic potentials whose basis functions are typically selected using a level-based scheme that is data-agnostic. We introduce a post-training, cost-aware pruning strategy that removes expensive basis functions with minimal loss of accuracy. Applied to nickel and silicon-oxygen systems, it yields models up to seven times faster than standard MTPs. The method requires no new data and remains fully compatible with current MTP implementations.

Machine-learning interatomic potentials (MLIPs) offer a tunable compromise between the accuracy of first-principles methods and the efficiency of empirical potentials. MLIP frameworks include neural network potentials, Bayesian approaches, and linear regression on systematically constructed basis functions. The latter category provides a practical balance between cost and accuracy for many materials, encompassing established MLIPs such as the Spectral Neighbor Analysis Potential (SNAP) [1], the Moment Tensor Potential (MTP) [2], and the Atomic Cluster Expansion (ACE) [3].

Efforts to improve the efficiency of MLIPs have been made on several fronts. On the training side, one major direction has been to reduce the cost of generating high-quality data, which depends on expensive quantum-mechanical calculations. Active learning strategies [4,5] and automated data-driven generation [6,7] have lowered the cost of curating effective training datasets. Some of our recent work has shown that smaller training configurations can capture essential physics while substantially reducing data-generation costs [8–11].

As machine-learning potentials are increasingly applied in large-scale Molecular Dynamics (MD), the computational cost of evaluating atomic energies and forces can become a limiting factor. On the hardware side, several efforts have focused on high-performance implementations using GPU acceleration through Kokkos-based backends in the LAMMPS [12] MD engine [13,14]. We recently contributed an efficient GPU-enabled MTP implementation in LAMMPS [15].

A complementary direction for improving efficiency lies in the algorithmic design of the basis set itself. For example, Thompson et al. used the DAKOTA optimization framework [16] to tune the hyperparameters in SNAP, such as the bispectrum components and cutoff

radius [1]. In the ACE potential, users specify hyperparameters such as the correlation order (ν), the maximum radial index ($n_{\max}$), and the maximum angular momentum ($l_{\max}$), which together define the basis. The basis can also be systematically expanded using a hierarchical ladder fitting scheme [14,17].

By contrast, the MTP basis is governed by a single level hyperparameter that counts combinations of radial and angular indices with fixed weights. This design is straightforward and systematic, but it implicitly assumes that all tensor contractions contribute equally to the balance between accuracy and computational cost. In practice, the basis functions are evaluated through a recursive style compute tree that is highly cost-asymmetric: different contractions can differ in cost by more than an order of magnitude. Recently, Wang et al. proposed alternative level definitions that increase the number of radial parameter sets, leading to improved performance [18].

Here, we introduce a post-training pruning strategy for the MTP basis that explicitly accounts for computational cost. Starting from a fully trained base potential, we apply a multiobjective evolutionary algorithm (EA) to identify subsets of basis functions that jointly optimize accuracy and cost. Accuracy is estimated by refitting the linear coefficients while keeping the radial components fixed, and cost is evaluated heuristically by traversing the compute tree. The procedure yields a population of potentials distributed along a cost-accuracy Pareto front. The resulting pruned MTPs can be used directly with existing implementations, including MLIP-2 [19], MLIP-3 [20], and our LAMMPS implementation [15], without code modification. Although

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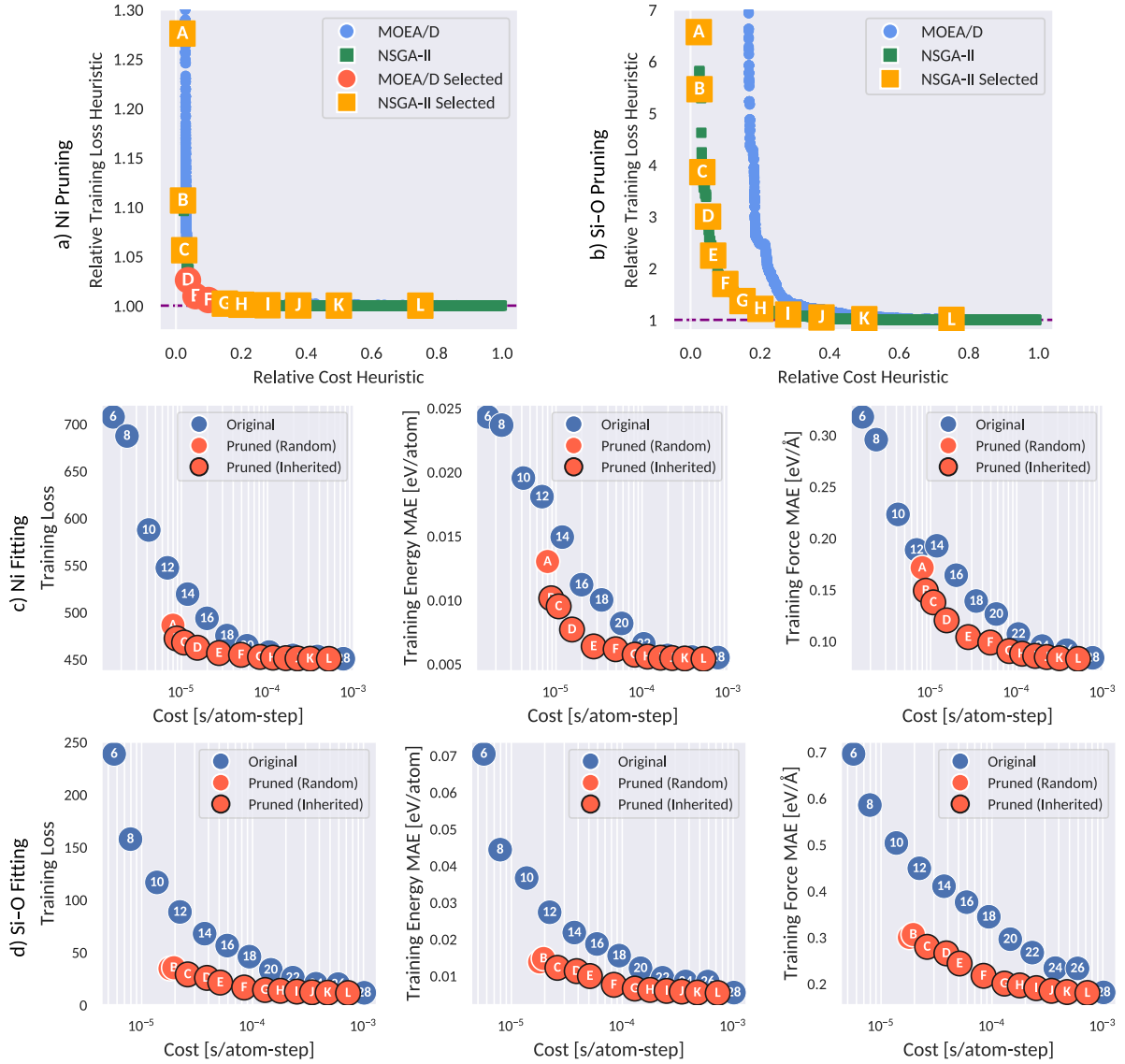


Fig. 1. The pruning results (a,b) show the Pareto fronts of *heuristics* obtained using NSGA-II and MOEA/D, relative to the base level-28 MTP. NSGA-II was generally more Pareto-optimal than MOEA/D. Twelve representative potentials (A–L) were selected to illustrate the range of achievable cost-accuracy tradeoffs, and fitted. **The fitting results (c,d)** include *realized* training loss, MAE energy per atom, and MAE force for Ni and Si–O. On one AMD EPYC 9654 core for 2048 FCC Ni and 1944 α -quartz atoms, the *realized* speedups are substantial: $3.8\times$ (Ni, MTP E) and $7.0\times$ (Si–O, MTP F). Inherited initialization typically delivered better losses than Random.

pruning was tuned for our CPU implementation, the resulting models remain compatible with GPU backends with potentially different scaling behavior. Additional details are provided in the Supporting Information.

The optimization workflow was implemented in Python using the pymoo framework [21], primarily employing the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [22], with limited tests using the Multiobjective Evolutionary Algorithm based on Decomposition (MOEA/D) [23]. Numerical operations relied on NumPy [24], and compute-tree traversal was accelerated with Numba [25]. Parallel evaluations were performed using mpi4py [26,27], replacing pymoo’s built-in parallel backend. While strong scaling remains bounded by serial components of the evolutionary algorithm, this approach enables efficient exploration of the Pareto front. The code is open-source and available as a pip-installable package from GitHub [28].

We demonstrate our pruning strategy on two representative systems: nickel, using the dataset of Andolina et al. [29] from Colab-Fit [30] as an example of a simple face-centered-cubic (FCC) metal, and silicon–oxygen, using the 1159-structure dataset of Zongo et al. [31] as

a complex covalent-ionic system. For both materials, reference MTPs were fitted using MLIP-3 across levels 6–28 following the standard level definitions of the MLIP code. Each level was fitted 32 times with randomized initial parameters, and the model with the lowest training loss was retained for further evaluation. The level-28 MTP served as the base potential for subsequent pruning. The resulting cost-accuracy Pareto fronts are shown in Fig. 1a and b, from which twelve representative potentials were selected, taking the better-performing results from either of the two evolutionary algorithms.

To assess how initialization affects the fitting of the pruned potentials, we compared two strategies: random initialization and parameter inheritance from the base level-28 MTP. In the random initialization strategy, the twelve representative MTPs (A–L) were fitted from scratch using MLIP-3, following the same random seeding protocol as for the level-based potentials. The inherited initialization refitted MTPs (A–L) with MLIP-3 using the base potential’s parameters as a starting point. Fig. 1c and d compares the better-performing initialization for each pruned model against the original MTP levels in terms of training loss, energy per atom mean absolute error (MAE), and force MAE. In both

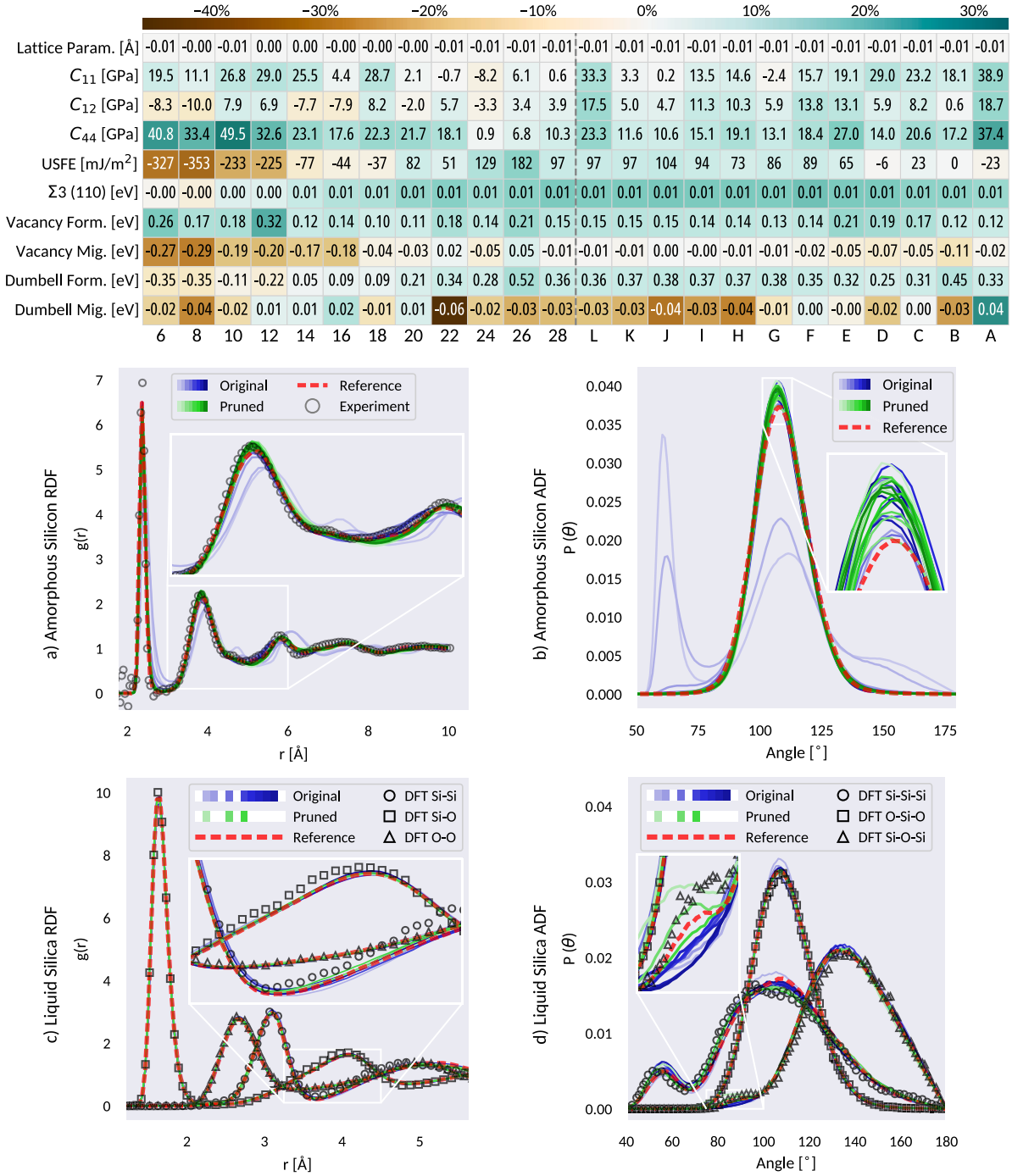


Fig. 2. Pruned MTPs generally reproduce the structural and energetic behavior of the highly expressive base level-28 MTP at much lower cost. **Ni errors (heatmap)** are presented relative to DFT, including lattice parameter; elastic constants C_{11} , C_{12} , and C_{44} at 300 K; USFE along $\langle 112 \rangle$ on $\{111\}$; $\Sigma 3(110)$ tilt grain boundary formation energy; and formation and migration energies for monovacancies and $\langle 110 \rangle$ dumbbell self-interstitials. **Amorphous Si structure data (a,b); Liquid SiO₂ structure data (c,d):** DFT results, and the reference level-26 MTP are from Zongo et al. [31]; experimental data is from Laaziri et al. [32]. Insets highlight regions of higher variability, such as b, where pruned potentials better capture the secondary bump as compared to the level-based potentials. Color luminance indicates MTPs within each group ordered by increasing cost; in silica, white denotes unstable potentials.

material systems, inherited initialization generally yielded faster convergence and lower training error, except for the lowest-cost models, where both initialization schemes performed comparably.

In nickel, model E achieves lower training loss and mean absolute errors for both energy and forces than the level-22 MTP, while running 3.8 $\times$ faster. In silicon–oxygen, model F attains better training accuracy than the level-26 MTP with a 7.0 $\times$ reduction in cost, and model A performs similarly to the level-20 MTP while being 8.1 $\times$ faster. The

speedups in Si–O are greater due to more opportunities for optimization. Nickel is a simpler system where we observed convergence in training loss around MTP level 22, in part due to 7 outlier configurations with small interatomic distances that the MTP radial basis construction had difficulty representing.

The structure of the pruned models clarifies the limitations of data-agnostic, level-based construction schemes. In Si–O, for instance, the level-12 MTP contains 127 learnable parameters, whereas potential A

is 22.9% faster despite using 386 parameters. The discrepancy arises from the allocation of learnable parameters rather than their count. The MTP includes per-neighbor and per-neighborhood costs, and Si–O has many neighbors on average. Thus, the pruning retains basis functions that reduce per-neighbor costs, whereas Ni, with only 44% of Si–O's average neighbors, benefits from retaining them. For instance, Potential L in both case studies achieves a $\approx 40\%$ speedup relative to their base potentials without compromising fit quality. However, in Ni, all per-neighbor costs are retained, while in Si–O, 17% are pruned. Additionally, all pruned potentials retained all the base potential's radial parameter sets, corroborating Wang et al.'s alternative level definition, which promotes more radial parameter sets.

To assess whether the cost-accuracy gains observed during pruning are reflected in physically meaningful quantities, we compared the pruned and level-based MTPs across a range of physical properties. The complete benchmark data and trained potentials are available from Zenodo [33]. In nickel, the benchmarking suite of Thoms et al. [34] was used for the heatmap of errors in Fig. 2, including the equilibrium lattice parameter, 300 K elastic constants, the unstable stacking fault energy (USFE) along $\langle 112 \rangle$ on $\{111\}$, the $\Sigma 3(110)$ tilt grain boundary formation energy, and the formation and migration energies of monovacancies and $\langle 110 \rangle$ dumbbell self-interstitials. In silicon-oxygen, the results were compared against Density Functional Theory (DFT) predictions and the level-26 MTP distributed alongside the Zongo et al. dataset [31]. This reference MTP employed different fitting hyperparameters targeting the accuracy of structures and was initialized randomly, with the best seed selected based on physical-property performance rather than training loss. Fig. 2 also shows the radial (RDF; a,c) and angular (ADF; b,d) distribution functions for amorphous silicon (a,b) and liquid silica (c,d).

Overall, the pruned models reproduce the behavior of the full level-28 potential remarkably well. This is especially true for models fitted with inherited initialization. In other words, pruning alters the computational workload far more than it alters the underlying physics captured by the fit. The main exception is the nickel elastic constants, which show slightly larger variations—a reminder that elastic responses are often among the most sensitive quantities in any interatomic potential.

In liquid silica at 3000 K, several potentials became unstable during molecular dynamics, an outcome that depended strongly on the initialization. By using different random seeds, both the level-based and pruned potentials could be stabilized, suggesting the instability of the pruned potentials may have been inherited from the unstable base potential. The instabilities likely arise from nonphysical behavior near the MTP's minimum distance. An additional short-range repulsive model, such as a Ziegler–Biersack–Littmark (ZBL) [35] or Nordlund–Lehtola–Hobler (NLH) [36] potential—together with MTP smoothing at the transition—would likely eliminate these issues. A more subtle example of inherited behavior appears in the Si–O–Si angular distribution of liquid silica (Fig. 2d, inset). Between 80° and 100° , DFT predicts a secondary bump absent in the original level-based potentials. Interestingly, every stable pruned potential reproduces this feature to some extent, implying that pruning preserved some information encoded in the more complex base potential or yielded stronger generalization.

Overall, our approach shows that substantial performance gains can be achieved in MLIPs by optimizing their structure. Even with our simple heuristics for estimating accuracy and cost, the resulting potentials offer strong speedups. In practice, pruning can be tuned to match specific simulation goals. For instance, the neighbor count used in our pruning was the average of the training set, but could easily be adapted to high-temperature or high-density conditions. Furthermore, pruning begins from a rich base potential; it compresses what exists rather than adding new terms. Starting from larger bases, such as MTPs with more radial parameter sets, would open even broader optimization spaces, although higher pruning costs would be incurred.

While we have focused on the MTP, a similar approach could be considered in other formalisms. ACE may be a good candidate for cost-aware optimization since it uses a similar recursive style evaluation of its basis functions.

From a practical standpoint, pruned MTPs can be used immediately in MLIP-3 and our own LAMMPS MTP module, requiring no code changes and yielding speedups of up to sevenfold. Faster potentials enable longer simulations, larger systems, and broader statistical sampling on the same hardware. Because our post-training method requires no new training data and is released as an open-source package with examples, it can be easily adopted by the community.

CRediT authorship contribution statement

Zijian Meng: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Karim Zongo:** Writing – review & editing, Validation. **Matthew Thoms:** Writing – review & editing, Validation. **Christopher Maxwell:** Writing – review & editing. **Edmanuel Torres:** Writing – review & editing. **Ryan Grant:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Laurent Karim Béland:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Gemini 2.5 Pro in order to check grammar, wording, and spelling; accelerate code prototyping, commenting, and refactoring (though all ideas and algorithms are the authors' own); write Python scripts to generate plots from the authors' own data; and correct $\LaTeX$ formatting and equation issues. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article, confirming that all intellectual and analytical contributions are solely those of the authors.

Data availability

All trained potentials and their validation data are available from Zenodo [33]. The latest pruning code and the version that produced the results herein are available from GitHub or Zenodo [28].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.commatsci.2026.114725>.

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